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Studierichting: Informatica
Oefeningenreeks 5E nr. 3

Bereken de oppervlakte ingesloten door de hypocycloïde met parameter-
voorstelling:

$$\begin{cases} x(t) = \frac{a}{3}(2 \cos t + \cos 2t) \\ y(t) = \frac{a}{3}(2 \sin t - \sin 2t) \end{cases}$$
$$S = \frac{1}{2} \left| \int_0^{2\pi} (x(t) \cdot y'(t) - y(t) \cdot x'(t)) dt \right|$$

afleiden geeft:

$$x'(t) = \frac{a}{3}(-2 \sin t - 2 \sin 2t) = -\frac{2a}{3}(\sin t + \sin 2t)$$

$$y'(t) = \frac{a}{3}(2 \cos t - 2 \cos 2t) = \frac{2a}{3}(\cos t - \cos 2t)$$

$$x \cdot y' - y \cdot x':$$

$$\begin{aligned} & \left[\frac{a}{3}(2 \cos t + \cos 2t) \cdot \frac{2a}{3}(\cos t - \cos 2t) \right] - \left[\frac{a}{3}(2 \sin t - \sin 2t) \cdot \left(-\frac{2a}{3} \right) (\sin t + \sin 2t) \right] \\ &= \frac{2a^2}{9} [(2 \cos^2 t - 2 \cos t \cos 2t + \cos 2t \cos t - \cos^2 2t) + (2 \sin^2 t + 2 \sin t \sin 2t - \sin 2t \sin t - \sin^2 2t)] \\ &= \frac{2a^2}{9} [2(\cos^2 t + \sin^2 t) - (\cos^2 2t + \sin^2 2t) - (\cos t \cos 2t - \sin t \sin 2t)] \\ &\Rightarrow \frac{2a^2}{9} [2(1) - (1) - \cos(t + 2t)] = \frac{2a^2}{9}(1 - \cos 3t) \end{aligned}$$

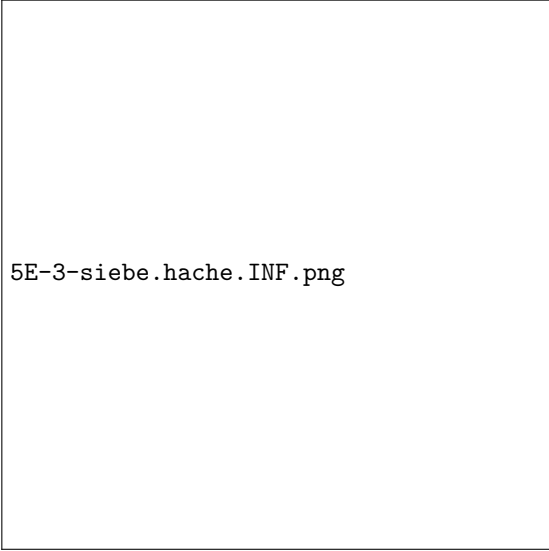
integraal:

$$S = \frac{1}{2} \int_0^{2\pi} \frac{2a^2}{9}(1 - \cos 3t) dt$$

$$S = \frac{a^2}{9} \int_0^{2\pi} (1 - \cos 3t) dt$$

$$S = \frac{a^2}{9} \left[t - \frac{1}{3} \sin 3t \right]_0^{2\pi}$$

$$S = \frac{a^2}{9} ((2\pi - 0) - (0 - 0)) = \frac{2\pi a^2}{9}$$



5E-3-siebe.hache.INF.png

References

- [1] Grafiek gemaakt met Python